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Table 1: Renderer vs. Training Pipeline Ablation, NTU-60 48/12 Xsub, identical pipeline (W1/Q1)

Renderer	Stage 1&4 Only	Full Four-Stage	Δ (Pipeline)
3D+Velocity	56.18	58.77	+2.59
DrAction (Ours)	62.09	64.72	+2.63
Δ (Renderer)	**+5.91**	**+5.95**	—

Table 2: Single-Person vs. Two-Person Actions, NTU-60 48/12 (W2/Q2)

Action Type	SkeletonLLM	TDSM	PURLS
Single-person	60.27	58.42	44.10
Two-person	75.37 (+15.10)	50.85 (-7.57)	34.23 (-9.87)

Table 3: Joint Dropout Robustness, NTU-60 48/12 Xsub, 3-run avg (W3/Q3)

Dropout	SkeletonLLM	InternVL3-8B† (Fixed)	TDSM
0%	64.72	56.28	56.03
10%	63.44 (-1.28)	53.58 (-2.70)	52.76 (-3.27)
20%	61.49 (-3.23)	49.89 (-6.39)	47.58 (-8.45)
30%	58.88 (-5.84)	45.02 (-11.26)	41.83 (-14.20)
50%	52.37 (-12.35)	37.34 (-18.94)	31.07 (-24.96)

Table 4: Gaussian Noise Robustness, NTU-60 48/12 Xsub, σ in cm, 3-run avg (W3/Q3)

σ (cm)	SkeletonLLM	InternVL3-8B† (Fixed)	TDSM
0	64.72	56.28	56.03
2	63.83 (-0.89)	54.79 (-1.49)	54.44 (-1.59)
5	61.71 (-3.01)	51.34 (-4.94)	49.87 (-6.16)
10	56.62 (-8.10)	44.71 (-11.57)	42.03 (-14.00)

Table 5: Modality Bridge Component Comparison, both ˜7–8B LLM (W4/Q4)

Component	Params	FLOPs
SUGAR: CTR-GCN	1.46M	2.0G
SUGAR: TQP (Q-Former)	188M	223.1G
Ours: DrAction	12.6K	33.2G

Table 6: Capability Comparison — SUGAR vs. SkeletonLLM (W4/Q4)

Capability	SUGAR	SkeletonLLM
Closed-set classification	✓	✓
Open-vocabulary recognition	Predefined list	✓
Cross-format zero-shot transfer	✗	✓
Captioning / QA / Causal reasoning	✗	✓

Figure 1: FPS Concept Illustration (W4/Q4)

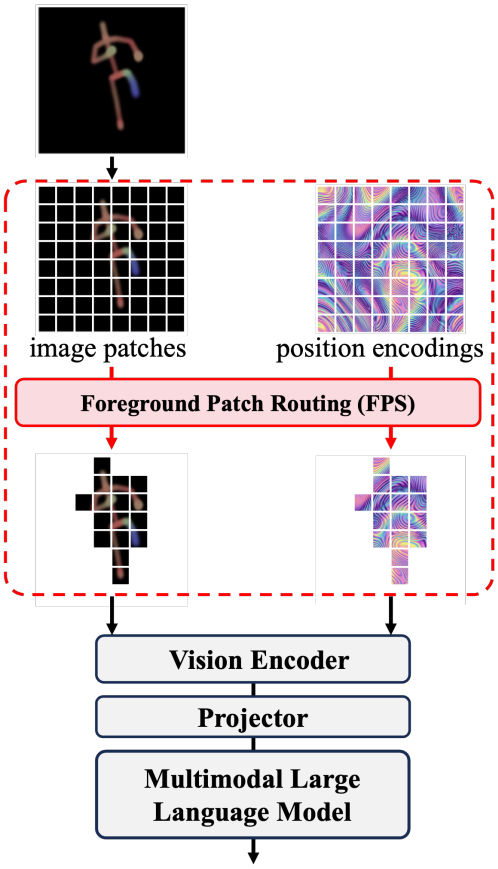

Table 7: Foreground Patch Sparsification — FPS, NTU-60 12-frame (W4/Q4)

Config	Resolution	Eff. Patches	Vision FLOPs	Tokens	55/5	48/12	40/20	30/30	**Avg**
Low-res	224×224	3,072	1.94T	768	83.12	59.88	41.27	32.69	54.24
Standard	448×448	12,288	8.67T	3,072	**87.37**	**64.72**	**46.15**	**37.84**	**59.02**
+ FPS	448×448	**2,244**	**1.41T**	**562**	85.75	63.24	44.96	35.80	57.48

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Table 8: Sec. 3 Revision Summary (W1, line numbers refer to camera-ready draft)

#	Issue	Revision	Lines
1	K_{bone}	Inline def. added	**202–204 (p.4, left)**
2	Quaternions	Scope clarified (primitives, not joints)	**L205-209, L221-223 (p.4, right → p.5, left)**
3	Canonical pos.	Inline def. added	**216–220 (p.4, right → p.5, left)**
4	Blend weights	Computation detailed	**225–232 (p.5, left)**
5	GRU "optional"	Design choice clarified (App. A.3)	**261–264 (p.5, left)**
6	Saliency gate	Linked to opacity formula	**266–270 (p.5, left)**
7	$\dot{\mathbf{j}}$, σ	Inline defs. added	**259–260, 272–274 (p.5, left)**
8	CE/BCE	New "MQA" paragraph	**245–255 (p.5, right); 287–290, 303–304, L310-311 (p.6, left)**
9	Fixed baseline	3D+Velocity, 448^2 specified	**326–329 (p.6, right); 332 (p.7, table caption)**
10	MLLM → classif.	Covered by #8	**245–255 (p.5, right)**

Table 9: Sec. 2.3 Citation Additions (W2)

Strategy	Added Citation
Joint remapping heuristics	Duan et al., 2022 (PySkl)
Zero-padding to max joint count	Wang et al., 2024
Format-specific adapters	Guo et al., 2023; Wang et al., 2025

Table 10: Single-Model Multi-Task Evaluation, same NTU-60 55/5 checkpoint, no task-specific retraining (W3/Q3)

Task	Test Set	Format (J)	Ref.	Superv.?	Metric
Open-vocab recog.	NTU-60 (55/5)	Kinect v2 (25)	Tab.1	✓	87.37% Acc
Cross-format recog.	NW-UCLA	Kinect v1 (20)	Tab.2	✗	60.38% Acc
Cross-format caption.	HumanML3D	SMPL (22)	Tab.3	✗	37.28 BertScore
Motion QA	Skeleton-QA	Kinect v2 (25)	Tab.15	✗	68.4 / 64.7% Acc

Table 11: DrAction + External MLLMs, NTU-60 zero-shot (W4/Q2)

Method	48/12	30/30
GPT-4o + 3D+Velocity	35.74	19.27
GPT-4o + DrAction	43.52	25.68
GPT-5.4 + DrAction	53.18	33.11
SkeletonLLM	**64.72**	**37.84**

Table 12: Cross-Dataset Joint Training, NTU-60 (25J) + HumanML3D (22J) (Q4)

Training	NTU-60 55/5	NTU→NW-UCLA	Skeleton-QA (Temp / Causal)
NTU-60 only	87.37	60.38	68.4 / 64.7
Joint training	88.52	65.04	73.2 / 70.9